

TARGET
CUSTOMER:

**Goat farmers, livestock farm owners, and dairy
producers who need efficient health and
location monitoring for their goats.**



HYPOTHESES

No. OF CUSTOMERS
VALIDATED WITH

PROBLEM

**Farmers cannot monitor goats continuously.
Diseases are detected late.
Goat theft and loss are common.
Manual tracking wastes time and labor.**



- ☐ 0
- ☐ <5
- ☐ <10
- ☐ <20
- ☐ >=20

CONSTRAINTS & BARRIERS

**Farmers may have limited technical knowledge.
Internet connectivity may be weak in rural areas.
Initial device cost sensitivity.
Device must be durable, waterproof, and animal-safe.**



- ☐ 0
- ☐ <5
- ☐ <10
- ☐ <20
- ☐ >=20

VALUE / BENEFITS

**Saves monitoring time.
Improves goat health through early detection.
Prevents loss and theft.
Increases farm productivity and profit.**



- ☐ 0
- ☐ <5
- ☐ <10
- ☐ <20
- ☐ >=20

SOLUTION & USABILITY

**Smart collar with sensors + GPS tracker.
Mobile app dashboard for real-time monitoring.
Simple interface usable by non-technical farmers.
Automatic alerts for abnormal health or movement.**



- ☐ 0
- ☐ <5
- ☐ <10
- ☐ <20
- ☐ >=20

PRICE

**Device: Affordable one-time cost suitable for small farmers.
App: Low monthly subscription.
Goal: Cheaper than hiring additional labor.**



- ☐ 0
- ☐ <5
- ☐ <10
- ☐ <20
- ☐ >=20

Forge Innovation Rubric

NECESSITY & SIGNIFICANCE 9/20	Real-world application scenarios 024	Wide problem incidence 024	High severity/impact of problem 036	User/Customer validated use-cases 036
NOVELTY & ORIGINALITY 8/20	Unique & differentiated solution concept 036	Holistic solution offering conceptually validated functionality 0246		(Futuristic) Potential to emerge as an enduring solution 048
CUSTOMER MOTIVATION 6/20	Customers have assigned highest priority & urgency 06	Customers have failed with alternative solutions 024	Customers have strong intent to co-create & standardise solution 026	High degree of intent to invest in the immediate term 04
STRONG VALUE PROPOSITION 7/20	Demonstrate functional/operational benefits 036	Significant gains measurable, quantified & validated 048	Value proposition proven with reasonable number of customers 06	
DEMONSTRATED PROOF OF VALUE 7/20	Field tested for utility, usability, integration & deployment constraints 0468	Strong indication of product adoption through successful field trials 036	Proven for ease-of adoption and accepted as permanent solution 036	

39/100

PROBLEM - CUSTOMER - PRODUCT <i>Getting 'em all right!</i> List all the beneficiaries who would benefit or gain from solving this problem. Select that beneficiary profiled in this canvas: Livestock Farm Owners ☒ Dairy Producers Livestock Researchers Goat farmers ☒ 06 How important to you is solving this problem among other priorities? Monitoring goats is a daily critical task. Health issues or theft directly cause financial loss, so solving this problem is a top priority for farmers. 06 Importance	Can you describe the problem in your own words exactly how you experience it? As a farmer, I cannot continuously monitor all my goats. Sometimes I notice illness too late, or goats go missing during grazing. I need a reliable system that alerts me instantly and helps me manage my herd efficiently. 01 Problem Description	What are the other constraints/challenges that have to be overcome or accommodated while solving the problem? Limited internet connectivity in rural areas Farmers limited technical knowledge Budget constraints Device must be durable,waterproof,animal-safe 04 Constraints/Barriers
Is this Beneficiary the RIGHT Customer? Customer-Motivation + faces the problem most severely, and stands to benefit the most Low Med High ☐ ☐ ☒ + aware of the problem and able to set clear expectations on gains/benefits ☐ ☐ ☒ + dissatisfied with existing solutions as they are complex, expensive, unusable, or ineffective ☐ ☐ ☒ Customer-Commitment + willing to describe solutions attempted and specify deployment/usability constraints ☐ ☐ ☒ + strong intent to co-create an effective solution ☐ ☒ ☐ + willing to define the evaluation/acceptance criteria for buying the product ☐ ☐ ☒ + indicates the preferred price range for buying a product ☐ ☒ ☐ PV-CD Progress Indicator No. of Customers 0<5<10<20>=20 Problem definition documented as is being experienced by the beneficiary ☐ ☐ ☒ ☐ Magnitude of the problem (PAIN) has been measured and validated with beneficiary ☐ ☒ ☐ ☐ Incidence/Occurrence of the problem (FREQUENCY) has been assessed and validated with beneficiary ☐ ☐ ☒ ☐ Challenges/limitations of current solutions attempted have been discussed/analysed ☐ ☒ ☐ ☐ Gaps/Challenges with existing commercial products have been analysed/documentated ☐ ☒ ☐ ☐ Adoption barriers have been discussed/documentated ☒ ☐ ☐ ☐		

05

Benefits

CHALLENGE STATEMENT Describe the challenge statement as it is faced in the real-world by the target beneficiaries. Goat farmers struggle to continuously monitor the health, movement, and safety of their goats, leading to delayed disease detection, animal loss, theft, and reduced farm productivity.		?	USE-CASE From the different scenarios in which the challenge is faced, describe the specific use-case you wish to target. Real-time monitoring of goats in semi-open and open grazing farms using a smart robotic + IoT system that tracks location and health parameters and sends alerts to farmers.		↓
GAP ANALYSIS Alternatives How does the target user solve the challenge today? Manual supervision,Physical tagging (ear tags),CCTV near sheds What are the other alternatives available in the market today?			TARGET USER For this use-case describe the profile of the target user that you expect to adopt/test and accept your MUP as a permanent solution. Manage 20–200 goats,Depend on livestock as primary income,Have limited labor support		
Gaps What are the gaps in the way the challenge is solved today? No real-time health tracking,Cannot monitor goats in open fields,High labor cost What are the gaps/shortcomings?		🍦	UTILITY/FEATURES Jobs Describe the specific job the target user wants to complete (get done)? Monitor goat health continuouslyTrack goat locationReceive alerts for abnormal activity Pains What are the pains to be relieved? Manual monitoring stressLate disease detectionTheft or missing goatsLabor dependency		👤
CONSTRAINTS What are the usability/ deployment constraints to be considered for the solution to be easy to use and to adopt? Must be simple to operateLow learning curveLightweight and animal-safeMinimal maintenance Usability Skills/ExpertiseTime ConsumptionPhysical/Personal RiskProduct TrainingChanges to habits/ProcessAccessibility Affordable cost Works in low-network areasLong battery life Durable and waterproof Deployment Total Cost of OwnershipMaintenance/ServiceInstallation/IntegrationResources/MaterialsTime ConsumptionOTHERS:			Outcomes What are the expected outcomes? How do you quantify them? Reduce livestock loss by 20–40%Reduce labor cost by 15–25%Faster disease detection (within hours instead of days) Gains What are the gains to be created? Time savingsReduced mortalityIncreased profitBetter herd management		
		🚧	REFERENCES Indicate other details or reference materials to know more about the user/use-case and the existing solutions/alternatives Research on IoT-based livestock monitoring systems		📄

Startup Name:

Date:

Version:

Team Members:

Pains to be relieved

Cost: What does your customer find too costly? (takes a lot of time, costs too much money, requires substantial efforts)

Feeling Bad: What makes your customer feel bad? (frustrations, annoyances, things that give them a headache)

Performance of Known Competition: How are current solutions underperforming for your customer? (lack of features, performance, malfunctioning)

Main Difficulties & Challenges: What are the main difficulties and challenges your customer encounters? (understanding how things work, difficulties getting things done, resistance)

Negative social consequences: What negative social consequences does your customer encounter or fear? (loss of face, power, trust, or status)

Risks: What risks does your customer fear? (financial, social, technical risks, or what could go awfully wrong)

Keeps Awake at Night: What's keeping your customer awake at night? (big issues, concerns, worries)

Common Mistakes: What common mistakes does your customer make? (usage mistakes)

Barriers: What barriers are keeping your customer from adopting solutions? (upfront investment costs, learning curve, resistance to change)

Customers Jobs-to-be-done

Functional Job (Perform, complete a specific task, solve a specific challenge)

Monitor goats' health and location continuously
Detect illness early
Prevent goat theft or loss
Reduce manual labor in herd management
Track feeding and movement patterns

Social Job (Gain power, looking good or status)

Be seen as a modern, tech-savvy farmer
Improve farm reputation in the community
Compete with larger farms using smart technology
Gain trust from buyers and investors

Emotional Job (aesthetics, feel good, security)

Feel secure knowing goats are safe
Reduce stress about disease or theft
Gain peace of mind through real-time alerts
Feel confident managing the farm efficiently

Basic Needs (Communication, nutrition, belonging, etc)

Ensure proper nutrition and health of goats
Maintain stable farm income
Protect livestock (livelihood security)
Improve communication through mobile monitoring system

Gains to be created

Savings: Which savings would make your customer happy? (time money and effort)

Outcome Expectations: What outcomes does your customer expect and what would go beyond his/her expectations? (quality level, more of something, less of something)

Current Solutions – Delight?: How do current solutions delight your customer? (specific features, performance, quality)

Life or Job – Easier?: What would make your customer's job or life easier? (flatter learning curve, more services, lower cost of ownership)

Positive Social Consequences: What positive social consequences does your customer desire? (makes them look good, increase in power, status)

What are customers looking for? (e.g. good design, guarantees, specific or more features)

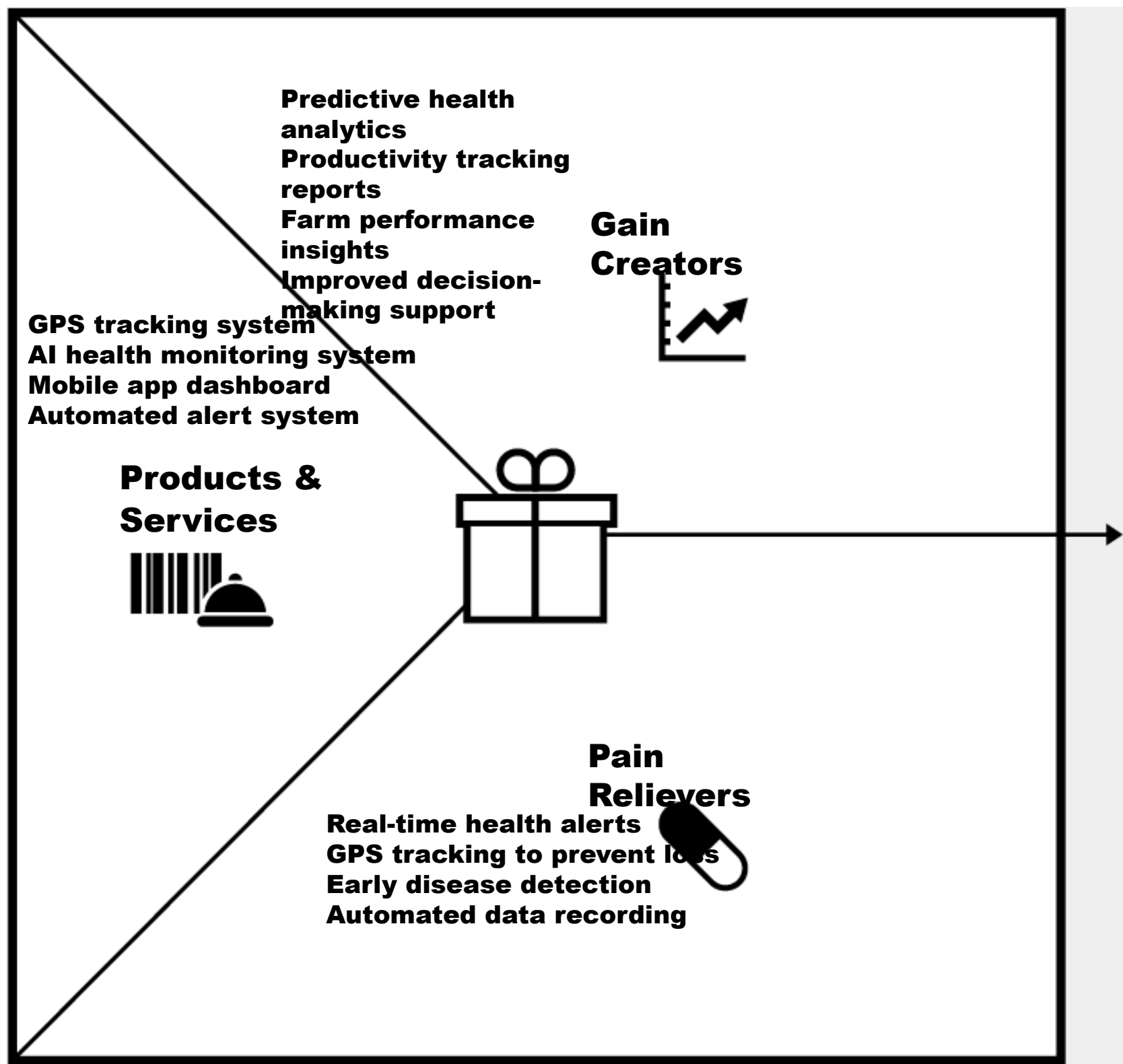
What do customers dream about? (e.g. big achievements, big reliefs)

How does your customer measure success and failure? (e.g. performance, cost)

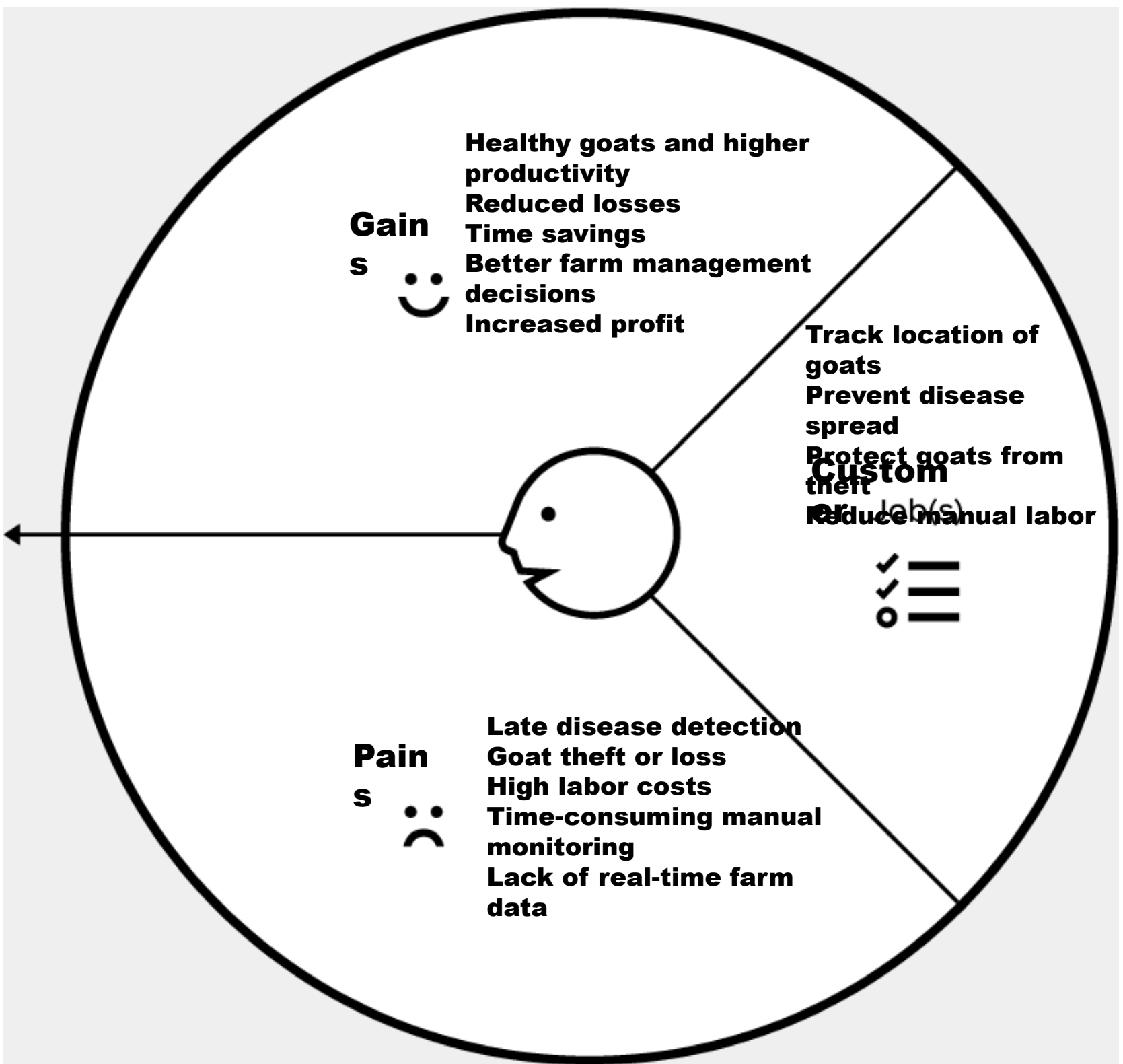
What would increase the likelihood of adopting a solution? (e.g. lower cost, less investments, lower risk, better quality, performance, design)

The Value Proposition Canvas

Value Proposition



Customer Segment





Our value proposition is
for (Target user)

Goat farmers and livestock farm owners.

who (Challenge statement)

who struggle to continuously monitor goat health, prevent disease outbreaks, and protect livestock from theft or loss.

the (Solution)

Goat Robotics

is a (Product category/theme)

smart IoT-based livestock monitoring and robotic farm management system.

it has (Compelling reason to buy)

real-time health tracking, GPS monitoring, early disease detection, and automated alerts that reduce labor and prevent losses.

unlike (Primary competitive alternatives)

manual monitoring, traditional tagging systems, and basic GPS trackers.

our Product (Primary differentiation)

integrates health sensors, AI analytics, and GPS tracking into one affordable, easy-to-use smart farming solution.



VALUE PROPOSITION

What are your Products & Services?
What are your Gain Creators?
What are your Pain Relievers?

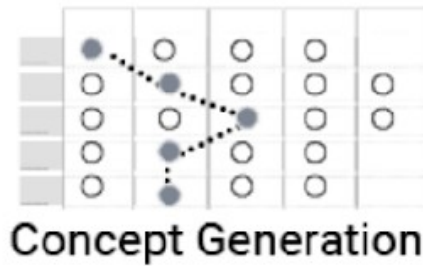
Smart IoT goat collar
GPS tracking system
Health monitoring sensors
AI analytics dashboard
Mobile app with alerts

ADOPTION BARRIERS

What are the constrains/challenges that have to be overcome or accommodated while solving the problem?

High initial device cost
Limited internet connectivity in rural areas
Farmers low technical knowledge
Maintenance and battery life concerns
Resistance to new technology

CONCEPT LEGOS



TOOLS/TECHNIQUES

FUNCTIONS	Option #1	Option #2	Option #3	Option #4	Option #5
Health Monitoring	Temperature sensor	Heart rate sensor	Motion/activity sensor	AI disease prediction	Vet alert integration
Location Tracking	GPS module	Geofencing alerts	GSM tracking	LoRa rural connectivity	RFID tagging backup
Data Management	Mobile app dashboard	Cloud storage	SMS alert system	Offline data sync	Farm analytics reports
Power System	Rechargeable battery	Solar-powered collar	Low-power IoT chipset	Battery health monitoring	Auto power-saving mode
Safety & Security	Theft alarm	Abnormal movement alert	Emergency panic alert	Night monitoring mode	Tamper detection

FUNCTIONS	CONCEPT #__	CONCEPT #__	CONCEPT #__
Track real-time GPS location Detect abnormal behavior Send instant alerts to farmer Store and analyze farm data Prevent theft using geofencing Reduce manual monitoring effort	Basic Smart GPS Collar Functions:	Advanced Health Monitoring Collar Functions:	Full AI Smart Farm System Functions:
Capability How feasible is the concept to build considering your capabilities? Usability How usable is the prototype overcoming adoption barriers to test the value proposition? Feasiblity How feasible is the solution to build considering the time and cost in building the prototype?	0 - 10 Capability 8 Usability 8 Feasiblity 9	0 - 10 Capability 7 Usability 9 Feasiblity 7	0 - 10 Capability 6 Usability 9 Feasiblity 6

USER/USE CASE Farmers use a smart collar system to monitor goat health, track location, and receive alerts for abnormal activity or theft.	PIR: Continuous monitoring of goats is difficult and time-consuming. How can we automate goat health and location monitoring in rural farms?	VALUE PROPOSITION: A smart IoT-based goat monitoring system that improves herd safety, health, and farm productivity with real-time alerts.	FEATURES AND FUNCTIONALITIES: GPS location tracking Temperature & activity monitoring Abnormal behavior detection SMS / App alerts Cloud data storage	ACTORS AND ROLES: Farmer – monitors goats Goat – data source (sensor carrier) UI/UX:															
SYSTEM CONTEXT: IoT collar attached to goat → Sends data via GSM/LoRa → Cloud server → Mobile app dashboard for farmer.	SYSTEM DESIGN: •UPS (Uninterruptible Power Supply) provides backup power when main electricity fails. •It stores energy in a battery and supplies power during outages. •The transformer regulates voltage and helps in charging and supplying power. •The circuit board controls the system and switches power automatically. •UPS protects devices from damage and ensures continuous power supply.																		
SYSTEM CONSTRAINTS: Rural network limitations Battery life constraints Cost sensitivity Weather durability requirement																			
INPUT: <table border="1"> <tr> <td>PHYSICAL QUANTITY:</td> <td>SENSOR/TRANSDUCER:</td> </tr> <tr> <td>Temperature, movement, location</td> <td>Temperature sensor, Accelerometer, GPS module</td> </tr> </table>		PHYSICAL QUANTITY:	SENSOR/TRANSDUCER:	Temperature, movement, location	Temperature sensor, Accelerometer, GPS module	<table border="1"> <tr> <td>INTERFACE :</td> <td> COMPUTE: HARD DISK MEMORY: Cloud database storage POWER: Rechargeable battery / Solar-assisted battery </td> <td>INTERFACE :</td> </tr> <tr> <td>GSM / WiFi / LoRa communication</td> <td></td> <td>GSM / WiFi / LoRa communication</td> </tr> </table>		INTERFACE :	COMPUTE: HARD DISK MEMORY: Cloud database storage POWER: Rechargeable battery / Solar-assisted battery	INTERFACE :	GSM / WiFi / LoRa communication		GSM / WiFi / LoRa communication	OUTPUT <table border="1"> <tr> <td>ACTUATOR :</td> <td>PHYSICAL QUANTITY:</td> </tr> <tr> <td>Buzzer / Mobile notification</td> <td>Alert message, health report, live location</td> </tr> </table>	ACTUATOR :	PHYSICAL QUANTITY:	Buzzer / Mobile notification	Alert message, health report, live location	HMI: Mobile application with dashboard, alerts, and farm analytics reports.
PHYSICAL QUANTITY:	SENSOR/TRANSDUCER:																		
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Startup Name:	Date:	Version:
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PROBLEM STATEMENT

Which problem do you aim to solve?

*Describe the problem as it is faced in the real world. If you are solving a technical problem then find a real-world product/service where this tech is currently applied, that's a good starting point.

Why is solving this problem very important?

Who are the different beneficiaries?

What goal can be achieved by solving this problem?

What gaps/challenges - in the way the problem is solved today, have to be overcome?

What are the different real-world scenarios where this problem is faced most critically?

Farmers struggle to monitor goat health and location continuously.

UNFAIR ADVANTAGE

In what other ways can your customer achieve the goal?

What other offerings in the market does your target customer compare your solution with now or is likely to do so in the future?

Who are your most important present and future competitors?

What serious business advantages do you enjoy over the competitors

What serious tech, operating or business advantages do you enjoy over your current or future competitors?

Integrated GPS + health monitoring in one affordable device.
AI-based early disease detection.
Rural-friendly design with low power consumption.
Simple mobile interface tailored for farmers.

TARGET CUSTOMER

Which beneficiary (type/category) gains most from solving this problem?

What should we look for if we have to find real-world customers of this beneficiary (type/category) in the real-world?

What does this beneficiary (type/category) expect as benefits/outcomes?

What would it be worth for this beneficiary (type/category) to solve this problem?

Primary: Goat farmers and livestock farm owners.
They expect better herd health, reduced losses, and time savings.
They are willing to invest if the solution reduces mortality and labor cost.

CUSTOMER ACQUISITION & SALES

How will you create awareness and generate leads for your product?

How do you increase the conversion from leads to trials, and from trials to sales?

Where would your customers actually buy your product from?

What is the channel for distribution and delivery of your product to the customer?

Awareness through agricultural exhibitions and farmer groups.
Partnership with veterinary clinics and farm cooperatives.
Sales via direct marketing, distributors, and online platforms.
After-sales support and demo trials to increase conversion.

PRODUCT

How are you solving this problem for your target customer?

What constraints faced by this customer are you overcoming with your solution?
*costs, resources, skills, operating conditions, time, convenience

How do you make this solution easy to adopt and use for your target customer?

Goat Robotics – a smart IoT collar with GPS and health sensors.
Provides real-time monitoring through a mobile app.
Designed to be easy to wear, durable, and simple to use.

PRICE & REVENUE MODEL

Who pays for your product? Who is the buyer & user? Are they the same?

For what value is your target customer willing to pay?

How much is your solution worth to your target customer?

Are other sources of revenue possible from other beneficiaries?

Revenue from hardware sales (smart collar).
Monthly subscription for app and analytics services.
Bulk pricing for large farms.
Additional revenue from advanced AI health reports.

VALUE PROPOSITION

Does your product help your target customer make or save money?

How is your product better than the alternative available today?

What extra benefits does the customer get from your product?

How do you make it easy for the customer to measure or experience the benefits?

Saves money by reducing goat loss and disease-related deaths.
Improves productivity with early health detection.
Better than manual monitoring or basic tags because it provides real-time alerts and analytics.
Benefits are measurable through reduced mortality and labor costs.

METRICS

Which outcomes should you measure to validate the assumptions in the creation & delivery of your core value proposition?

Are you maximising the conversion rate in the Get-Keep-Grow pipeline for your target customers?

Number of active users.
Reduction in goat mortality rate.
Customer retention rate.
Monthly recurring revenue (MRR).
Customer satisfaction and referrals.